Chess Fart

Version 1.0 Documentation



Check. Mate. Ventilate.

Chess Fart is a deliberately ridiculous 256-color VGA strategy game: real chess rebuilt as a loud early-1990s DOS game where pieces build Gas and use directional fart blasts as tactical abilities.

Design goal: strategy first, toilet humor second.

Inside This Manual

Gameplay Basics
Learn the board, HUD, menus, controls, and core game modes.

Fart Mode & Attract Demo
Understand Gas, directional Fart attacks, push mechanics, and the built-in demonstration.

Themes & Web Interface
See Royal Basement, Crimson Cellar, and the responsive browser/WASM version.

Build, QA & Release Status
Review the platforms, automated verification, compatibility scope, and final release status.

1. Gameplay Basics

Core Board Layout, HUD, and Menu Flow



Chess Fart follows the rules of standard chess while adding Gas and Fart abilities on top of the normal game.

The main game screen uses a native 320×200 VGA layout. The chessboard occupies the left side of the screen, while a compact HUD on the right shows the current turn, game state, selected piece, Gas level, mode, and recent action.

A command strip along the bottom provides the most important keyboard shortcuts without taking attention away from the board.

Chess Board



The 8×8 board shows the current position.

When a piece is selected, legal destination squares are highlighted. The cursor can be moved with the arrow keys, and pieces can be selected or confirmed with Enter or Space.

HUD



The right-side HUD displays:

- **Turn** — White or Black.
- **State** — Playing, Check, Checkmate, Stalemate, or another terminal state.
- **Selected Piece** — Piece type and square.
- **Gas** — Current Gas charges from 0–3.
- **Mode** — CPU, Local 2P, Practice, or a special action state.
- **Last Action** — Recent move, Fart action, or status message.

Title Screen



The title screen contains the following choices:

- **PLAY CPU** — Play against the computer.
- **2 PLAYERS** — Local hot-seat two-player chess.
- **PRACTICE** — Experiment with positions and use Undo.
- **ATTRACT DEMO** — Watch a short demonstration of Chess Fart's signature mechanic.
- **HELP / RULES** — Open the in-game help pages.
- **CREDITS** — View the credits.
- **QUIT TO DOS** — Exit the game.

2. FART MODE & ATTRACT DEMO

The Signature Mechanic



Fart Mode is the defining Chess Fart mechanic.

Pieces accumulate Gas. A piece with enough Gas can enter Fart Mode and attempt a directional blast.

The Fart mechanic does not replace chess. Standard chess legality still applies, including king safety.

Entering Fart Mode



Select a piece with available Gas and press:

F — Fart Mode

Choose one of eight directions using the arrow keys or keypad diagonals. The game previews the target direction and potential displacement before the action is committed.

Enter / Space — Confirm
Esc — Cancel

Gas



Gas is stored per piece.

The HUD displays each selected piece's current Gas level from:

0 / 3 to 3 / 3

Gas is a limited tactical resource and must be spent carefully.

Fart Pushes



A successful Fart can push an affected piece by one square. The result must remain legal under normal chess rules.

Important consequences include:

- A push can change tactical relationships.
- A pushed pawn can promote.
- A king may be pushed only if the resulting position remains legal.
- Moving a king or rook from its home square by a Fart can eliminate castling rights.
- A Fart may fail when the target direction is blocked or otherwise illegal.

Attract Demo



The title screen includes a deterministic Attract Demo that demonstrates a Fart push.

The certified sequence begins with:

- White knight on C3
- Black pawn on D4
- Charged knight ready to Fart
- Direction: NE

The preview displays:

AIM NE TO PUSH

The resulting action pushes the pawn:

D4 → E5

The final game message reads:

D4 BLASTED TO E5

The demo then completes and returns control to the title screen.

3. Themes & Web Interface

Royal Basement



Royal Basement

Royal Basement is the default visual theme.
Its presentation uses:

- Stone-grain board surfaces
- Classic gilt piece accents
- Dark blue VGA interface panels
- Gold, teal, and cream semantic colors

Royal Basement remains the startup and fallback theme.

Crimson Cellar



Crimson Cellar

Crimson Cellar is the alternate presentation theme.
It uses:

- Burgundy and copper coloring
- Cellar-brick board surfaces
- Copper piece accents
- The same gameplay geometry and sprite masks as Royal Basement

The theme affects presentation only.
It does not change:

- Chess rules
- Gas or Fart behavior
- CPU behavior
- Save state
- Replay data
- Match results

Theme Selection



Press T on the title screen to switch between:

Royal Basement ⟷ Crimson Cellar

The selected theme is persisted through the version-2 configuration file.

Browser / WebAssembly Edition



Chess Fart also builds as a WebAssembly browser edition.
The game itself still renders its authentic native 320×200 VGA framebuffer, while the surrounding browser shell adapts to modern displays.

Desktop



On a real desktop fine-pointer environment:

- Mouse and keyboard controls are available.
- Desktop action buttons remain visible.
- Keyboard help remains visible.
- The touch D-pad is hidden.

Tablet & Mobile



On touch-oriented layouts:

- The browser shell becomes narrower.
- Touch controls become available.
- The VGA game remains correctly scaled without changing its native aspect ratio.

The responsive shell has been validated at:

- 1365×900 desktop
- 768×1024 tablet
- 390×844 mobile
- 320×568 compact mobile

4. Build, QA & Release Status

Version 1.0 Is Feature Complete



Chess Fart 1.0.0 is complete and maintenance-only.

The gameplay feature set is frozen. Future maintenance is limited to defects, compatibility work, test hardening, documentation, and simplification.

Frozen Compatibility Contracts



Game save: v2
Replay file: v1
Configuration: v2

The gameplay rules and persisted data contracts are frozen for version 1.0.

Supported Build Targets



Host / Regression

Primary automated regression target.

Command: `make test`

The host gate uses strict C89-compatible compilation with warnings treated as errors.



WebAssembly

Browser edition built with Emscripten.

Command: `make -f Makefile.web web`



16-Bit DOS

The DOS edition is compiled with Open Watcom 2.0.

Command: `sh scripts/build_dos_ci.sh`

The automated DOS target produces a real 16-bit:

`CHESSFRT.EXE`

Release Packaging



Command: `sh scripts/package_release.sh`

The release package contains:

- **CHESSFRT.EXE**
- **DOSBOX.CONF**
- **README.TXT**
- **KEYS.TXT**
- **KNOWNISS.TXT**
- **BUILDINFO.TXT**

The package is integrity checked and accompanied by SHA-256 metadata.

5. QUALITY & VERIFICATION

Automated Verification

The release gate includes permanent regression coverage for:

- Standard chess rules
- Gas mechanics
- Fart displacement
- CPU search and tactics
- Persistence
- Local two-player
- Practice / Undo
- Replay timeline
- Replay export / import
- Audio
- UX and hit testing
- Terminal states
- Promotion
- Fart-driven promotion
- Save / Load rollback
- Theme persistence



Browser Verification

The current Web build passes:

- Browser runtime smoke — PASS
- Chromium gameplay playtest — PASS
- Deep Chromium UX audit — PASS
- Real desktop fine-pointer audit — PASS
- 48-state visual regression suite — PASS

The canonical visual suite currently contains 48 native 320×200 states. It covers:

- Title screens
- Normal gameplay
- Selected pieces and legal moves
- Fart Mode
- Valid and invalid Fart actions
- Promotion
- Check
- Checkmate
- Stalemate
- History
- Help
- Credits
- Replay
- Royal Basement
- Crimson Cellar
- Attract Demo

All new canonical states use pinned visual signatures to detect unintended rendering changes.



DOS Verification

Automated DOS evidence includes:

- Real Open Watcom 2.0 16-bit compilation
- DOSBox execution of the DOS smoke program
- VGA Mode 13h initialization
- INT 33h mouse initialization probe
- Audio initialization
- Release ZIP validation

This automated smoke coverage should not be described as a complete interactive DOS hardware certification.



Current Repository Status

Release CI: PASS
Web Build: PASS
GitHub Pages deployment: PASS
Open issues: 0
Open pull requests: 0



6. COMPATIBILITY & RELEASE CLOSEOUT

Verified

The following areas have automated evidence:



- Strict C89 host build
- Rules and gameplay regressions
- Browser / WebAssembly runtime
- Chromium gameplay flows
- Real desktop fine-pointer behavior
- Open Watcom 2.0 16-bit DOS compilation
- DOSBox platform smoke
- VGA initialization
- Mouse initialization probe
- Audio initialization
- Release ZIP structure and integrity

Manual / Community Compatibility Targets

The following remain deliberately unverified for 1.0:



- Physical 386DX / 486 VGA systems
- MS-DOS 6.x
- FreeDOS
- Real Sound Blaster 1.x / 2.0 / Pro / 16 hardware
- Microsoft-compatible DOS mouse-driver combinations
- DOSBox-X interactive sessions
- DOSBox Staging interactive sessions
- Full human EASY / MED / HARD matches on slow DOS-class CPU profiles

These are compatibility-expansion targets, not known release defects.

What Is Actually Left

The game itself is finished.
The remaining work is release presentation and repository housekeeping:



Documentation Cleanup

- Update older documentation references that still mention 36 canonical visual states so they reflect the current 48-state suite.
- Refresh old Build 17 visual references where appropriate.

Publish the GitHub Release

Create the official:

- **v1.0.0**
- Git tag and GitHub Release.
- Attach or link the verified DOS package and checksum, and include release notes describing the final game.

Until this publication step is complete, the correct status is:

Game Complete — Release Publication Pending
—not "GitHub release published."

Final Project Status

- Core development: COMPLETE
- Gameplay: COMPLETE
- UX cleanup: COMPLETE
- Browser qualification: COMPLETE
- DOS automated qualification: COMPLETE
- 48-state visual certification: COMPLETE
- Open issues: 0
- Open pull requests: 0
- GitHub v1.0.0 release publication: PENDING



Chess Fart has reached shipped-quality software status. Remaining work is documentation cleanup and formal publication rather than additional game development.

CHESS FART 1.0.0

Check. Mate. Ventilate.

A real chess game.
With Gas.
And consequences.